

Unit 2: Boolean Algebra

What you need to know: Boolean algebra is the math of digital logic. Unlike regular math that uses numbers like 2, 5, or 100, Boolean algebra only uses two values: **1 (True / On)** and **0 (False / Off)**.

This is exactly how your computer's brain processes information and makes decisions.

Here are the basic logic "gates" (the building blocks of computer chips) you need to understand:

- **Boolean Operations** These are the basic logical questions the computer asks to process data. Instead of adding or multiplying, the computer checks if conditions are True or False.
- **AND Gate**
 - **How to remember it:** The strict gate.
 - **The Rule:** The output is True (1) **only** if ALL inputs are True (1). If even one input is False (0), the entire output becomes False (0).
- **OR Gate**
 - **How to remember it:** The relaxed gate.
 - **The Rule:** The output is True (1) if **at least one** input is True (1). The only way to get a False (0) output is if every single input is False.
- **NOT Gate**
 - **How to remember it:** The flipper or inverter.

- **The Rule:** It simply reverses whatever you put into it. If the input is 1, the output is 0. If the input is 0, the output is 1.

- **NAND Gate**

- **How to remember it:** NOT + AND.
- **The Rule:** It does the exact opposite of the AND gate. It outputs False (0) *only* when all inputs are True. For everything else, it outputs True (1).

- **NOR Gate**

- **How to remember it:** NOT + OR.
- **The Rule:** It does the exact opposite of the OR gate. It outputs True (1) *only* when all inputs are False.

- **XOR Gate (Exclusive OR)**

- **How to remember it:** The "differences" gate.
- **The Rule:** It outputs True (1) only when the inputs are **different** from each other (like a 1 and a 0). If the inputs are exactly the same (both 1s or both 0s), it outputs False (0).

- **Truth Tables**

- **What it is:** A truth table is simply a handy chart you will draw to map out all the possible inputs (0s and 1s) and show what the final output will be for a specific gate. Whenever you are tested on logic gates, you will usually use a truth table to prove your answer.